

Macroeconomics #2

The Solow–Swan Growth Model

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Today's lecture

- ① Kaldor's Facts
- ② Basic Solow
- ③ Some Extensions of the Solow Growth Models

Why Do Countries Grow?

Why Are Some Countries Rich and Others Poor?

The big questions

- What determines the long-run standard of living?
- What is the role of savings, population growth, and technology?
- How does growth evolve over time—speed up, slow down, stabilize?
- Are there forces that allow poorer countries to catch up with richer ones?

Income differences across countries

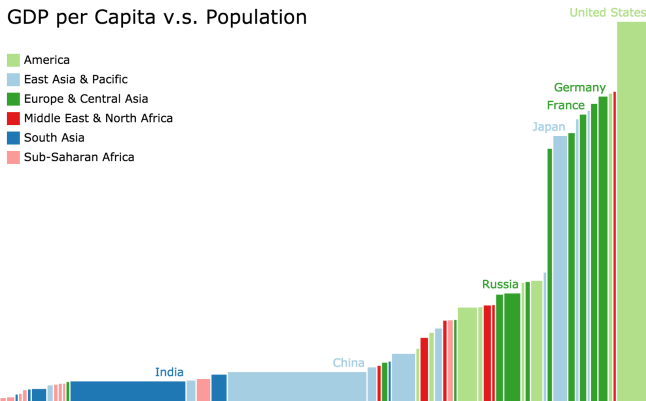
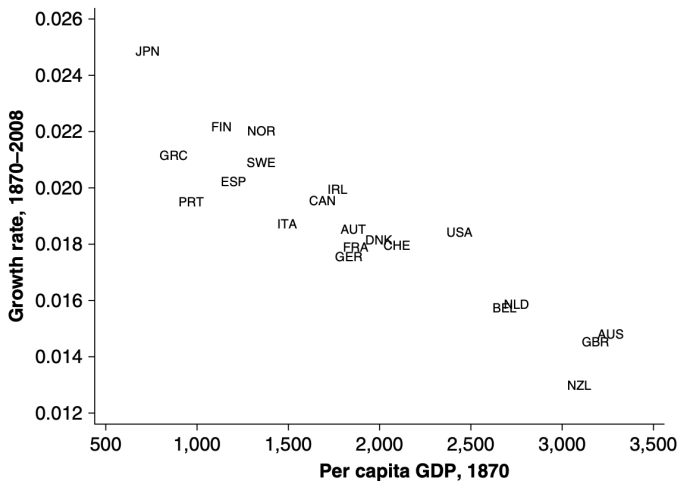


Figure: Distribution of Population and GDP per capita. Source: https://ilfat-galiev.im/docs/charticulator/gallery/gdp_vs_population

Convergence



SOURCE: Maddison (2010)

Figure: OECD growth rate versus per capita income in 1870. Lower initial

The Great Divergence

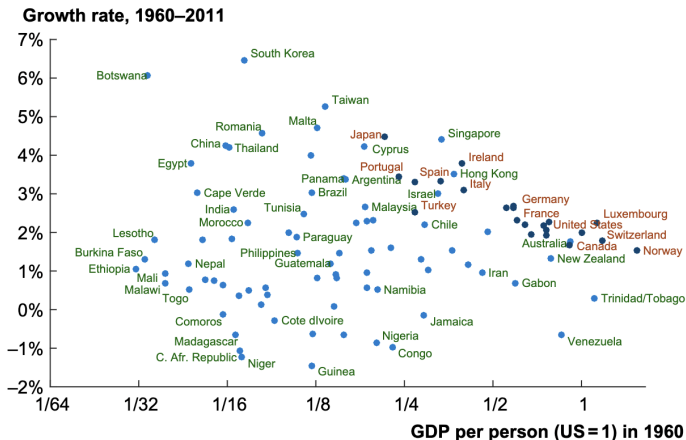


Fig. 26 The lack of convergence worldwide. Source: *The Penn World Tables 8.0*.

Kaldor's stylized facts of economic growth

Kaldor (1957) summarized the statistical properties of long-term growth:

- **SF 1:** Output per worker grows over time with *no tendency for a falling rate of growth of productivity*.
- **SF 2:** Capital per worker shows continuing growth.
- **SF 3:** The rate of return on capital is roughly constant.
- **SF 4:** The capital-output ratio is roughly constant.
- **SF 5:** Labour and capital receive constant shares of total income.
- **SF 6:** There are wide differences in the rate of productivity growth across countries.

Not all are independent: (SF 1) and (SF 4) imply (SF 2); (SF 4) and (SF 5) imply (SF 3).

US Real GDP per capita

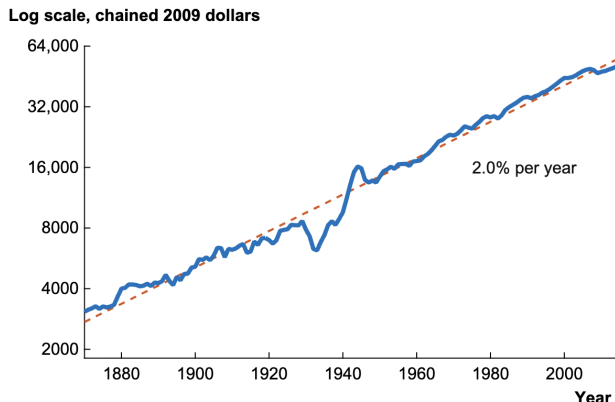


Fig. 1 GDP per person in the United States. Source: Data for 1929–2014 are from the U.S. Bureau of Economic Analysis, NIPA table 7.1. Data before 1929 are spliced from Maddison, A. 2008. *Statistics on world population, GDP and per capita GDP, 1-2006 AD*. Downloaded on December 4, 2008 from <http://www.ggdc.net/maddison/>.

Real GDP per capita growth is roughly constant over long horizons.

US Capital/Real GDP share

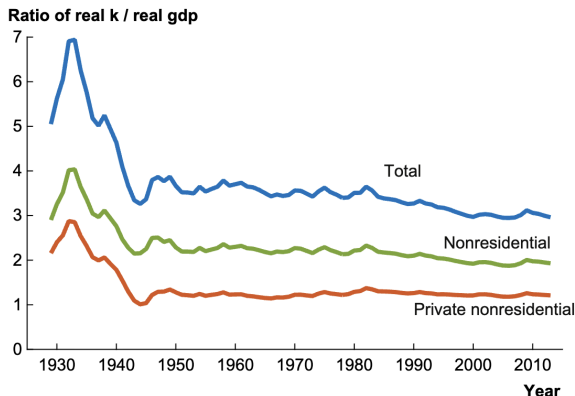


Fig. 3 The ratio of physical capital to GDP. Source: *Bureau of Economic Analysis Fixed Assets tables 1.1 and 1.2*. The numerator in each case is a different measure of the real stock of physical capital, while the denominator is real GDP.

Real capital / GDP is roughly constant over long horizons.

Labor share is roughly constant

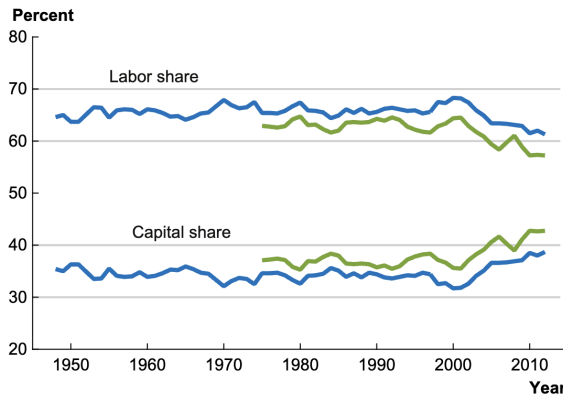


Fig. 6 Capital and labor shares of factor payments, United States. Source: *The series starting in 1975 are from Karabarbounis, L., Neiman, B. 2014. The global decline of the labor share. Q. J. Econ. 129 (1), 61–103. <http://ideas.repec.org/a/oup/qjecon/v129y2014i1p61-103.html> and measure the factor shares for the corporate sector, which the authors argue is helpful in eliminating issues related to self-employment. The series starting in 1948 is from the Bureau of Labor Statistics Multifactor Productivity Trends, August 21, 2014, for the private business sector. The factor shares add to 100%.*

The Basic Solow Model

(without technological progress)

Key features of the model

The Solow–Swan model is named after Robert Solow and Trevor Swan (1956).

- Closed economy, single final good.
- Continuous time ($t \in [0, \infty)$); we will also cover the discrete-time version.
- Large number of identical households.
- Households are *not* optimizing: they save a **constant fraction** s of income.
- We focus on the production side of the economy.

The model explains long-run regularities with minimal microeconomic structure.

Production

Aggregate production function:

$$Y = F(K, L)$$

- K : aggregate physical capital
- L : aggregate labor
- Y : aggregate output

Assumption 1 (Neoclassical): F is twice continuously differentiable, **constant returns to scale** in K and L , and satisfies

$$F_i > 0, \quad F_{ii} < 0, \quad i \in \{K, L\}$$

Assumption 2 (Inada conditions):

$$\lim_{K \rightarrow 0} F_K = +\infty, \quad \lim_{K \rightarrow \infty} F_K = 0, \quad \lim_{L \rightarrow 0} F_L = +\infty, \quad \lim_{L \rightarrow \infty} F_L = 0$$

Constant returns to scale (Review)

Definition

A function $P : \mathbb{R}^{n+k} \rightarrow \mathbb{R}$ is **homogeneous of degree m** if

$$P(\lambda x_1, \dots, \lambda x_n; z) = \lambda^m P(x_1, \dots, x_n; z) \quad \forall \lambda \in \mathbb{R}_+$$

When $m = 1$, we say P exhibits **constant returns to scale** (CRS, linearly homogeneous).

Theorem (Euler's Theorem)

If P is homogeneous of degree m , then

$$mP(x_1, \dots, x_n; z) = \sum_{i=1}^n x_i P_{x_i}(x_1, \dots, x_n; z)$$

For CRS production: $F(K, L) = F_K \cdot K + F_L \cdot L$, i.e., output is exhausted by paying each factor its marginal product.

Per worker variables

Use CRS: set $\lambda = 1/L$ to define **per worker** (intensive form) variables:

$$y \equiv \frac{Y}{L}, \quad k \equiv \frac{K}{L}$$

Then:

$$y = \frac{Y}{L} = F\left(\frac{K}{L}, 1\right) = f(k)$$

The per worker production function $f(k)$ inherits:

- $f(0) = 0$, $f'(k) > 0$, $f''(k) < 0$ (concavity, diminishing returns)
- $\lim_{k \rightarrow 0} f'(k) = +\infty$, $\lim_{k \rightarrow \infty} f'(k) = 0$ (Inada)

Example: Cobb–Douglas $F(K, L) = K^\alpha L^{1-\alpha}$ gives $f(k) = k^\alpha$, $\alpha \in (0, 1)$.

Households and savings

- Households save a **constant fraction** $s \in (0, 1)$ of income and consume the rest:

$$C = (1 - s)Y, \quad S = sY$$

- In equilibrium (closed economy, no government): $I = S = sY$.
- Population (labor force) grows at constant rate $n > 0$:

$$\frac{\dot{L}}{L} = n \quad \Rightarrow \quad L(t) = L_0 e^{nt}$$

Capital accumulation

Aggregate capital evolves as:

$$\dot{K} = I - \delta K = sY - \delta K$$

where $\delta \in (0, 1)$ is the depreciation rate.

Now derive the per worker law of motion. Since $k = K/L$:

$$\dot{k} = \frac{\dot{K}L - K\dot{L}}{L^2} = \frac{\dot{K}}{L} - \frac{\dot{L}}{L} \frac{K}{L} = \frac{\dot{K}}{L} - nk$$

Substituting $\dot{K} = sY - \delta K$:

$$\dot{k} = \frac{sY - \delta K}{L} - nk = sf(k) - \delta k - nk$$

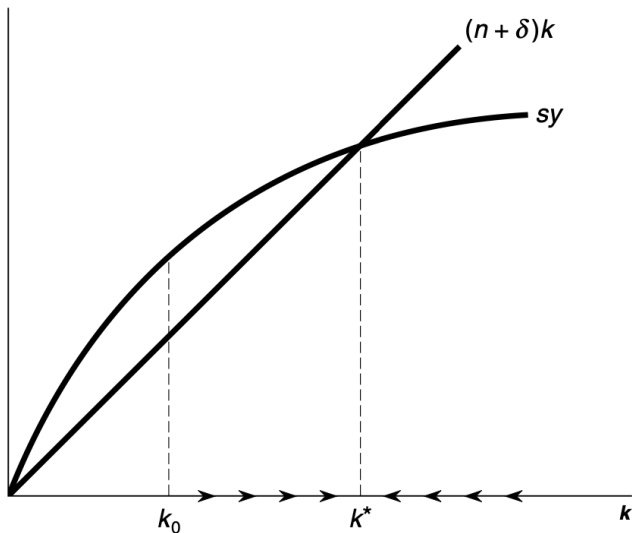
The fundamental equation of the Solow model

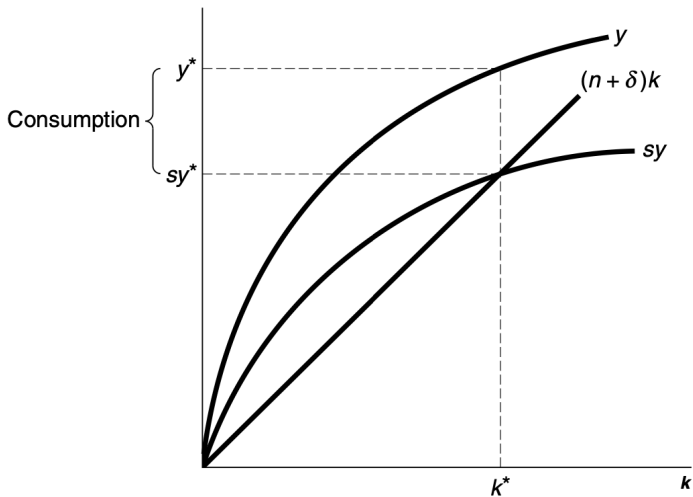
Solow equation

$$\dot{k} = sf(k) - (n + \delta)k$$

- $sf(k)$: actual investment per worker
- $(n + \delta)k$: break-even investment (investment needed to keep k constant)
 - δk : replaces depreciated capital
 - nk : equips new workers with capital
- $\dot{k} > 0$ when $sf(k) > (n + \delta)k$ (capital deepening)
- $\dot{k} < 0$ when $sf(k) < (n + \delta)k$ (capital shallowing)

Graphical analysis: the Solow diagram





Steady state

The steady state k^* solves $\dot{k} = 0$:

$$sf(k^*) = (n + \delta)k^*$$

For Cobb–Douglas $f(k) = k^\alpha$:

$$sk^{*\alpha} = (n + \delta)k^* \quad \Rightarrow \quad k^* = \left(\frac{s}{n + \delta} \right)^{\frac{1}{1-\alpha}}$$

Steady-state output per worker:

$$y^* = (k^*)^\alpha = \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

Steady-state consumption per worker:

$$c^* = (1 - s)y^* = (1 - s) \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

Stability of the steady state

Define $g(k) \equiv sf(k) - (n + \delta)k$. Then $\dot{k} = g(k)$ and:

$$g'(k) = sf'(k) - (n + \delta)$$

At the steady state:

- $g(k^*) = 0$
- $g'(k^*) = sf'(k^*) - (n + \delta) < 0$ (since $f'' < 0$ and Inada conditions)

By the linearization result from Lecture #1, k^* is **locally asymptotically stable**. In fact, under the Inada conditions, k^* is **globally stable**: from any $k_0 > 0$, the economy converges to k^* .

- If $k_0 < k^*$: $sf(k_0) > (n + \delta)k_0$ so $\dot{k} > 0$ (capital deepening)
- If $k_0 > k^*$: $sf(k_0) < (n + \delta)k_0$ so $\dot{k} < 0$ (capital shallowing)

Growth rate of k

The growth rate of capital per worker:

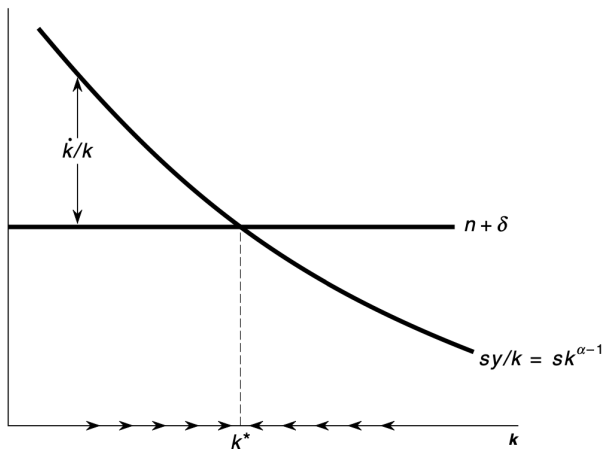
$$\frac{\dot{k}}{k} = s \frac{f(k)}{k} - (n + \delta) = sk^{\alpha-1} - (n + \delta)$$

Since $\alpha - 1 < 0$, the function $sk^{\alpha-1}$ is **decreasing** in k :

- When k is small, the growth rate is high.
- As $k \rightarrow k^*$, the growth rate falls to zero.

The growth rate of output per worker: $\frac{\dot{y}}{y} = \alpha \frac{\dot{k}}{k}$.

Transition dynamics



Key implication: countries far below k^* grow fast; growth slows as they approach steady state.

Comparative Statics

Effect of the savings rate

An increase in s shifts $sf(k)$ upward:

$$k^* = \left(\frac{s}{n + \delta} \right)^{\frac{1}{1-\alpha}} \Rightarrow \frac{\partial k^*}{\partial s} > 0$$

- Higher s raises steady-state k^* and y^* .
- But s has a **level effect**, not a **growth effect**: the long-run growth rate of y is still zero (without technology).
- During the **transition**, higher s temporarily raises the growth rate.

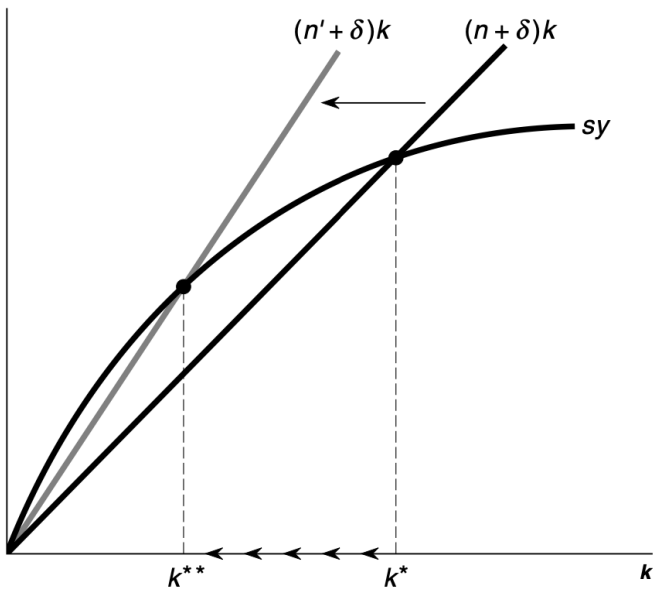
Effect of population growth and depreciation

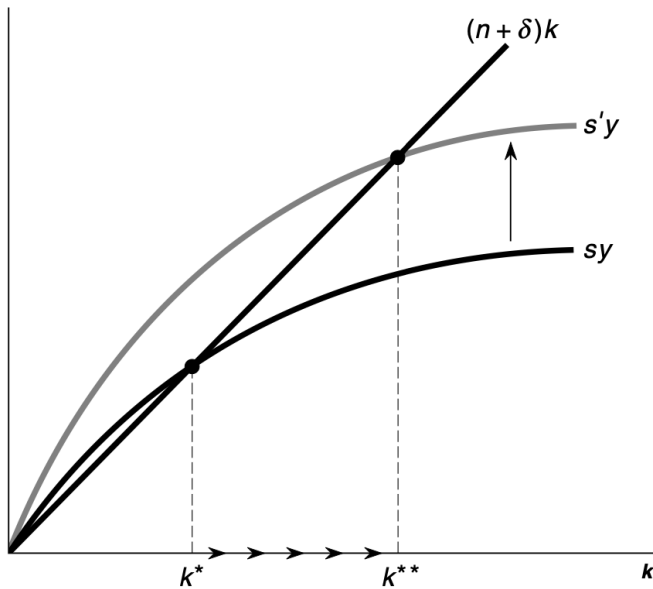
An increase in n or δ rotates the break-even line $(n + \delta)k$ upward:

$$\frac{\partial k^*}{\partial n} < 0, \quad \frac{\partial k^*}{\partial \delta} < 0$$

- Higher n : more workers to equip $\Rightarrow$ lower k^* , lower y^* .
- Higher δ : more capital wears out $\Rightarrow$ lower k^* , lower y^* .

The model predicts: countries with higher savings rates are richer; countries with higher population growth rates are poorer. This is broadly consistent with the data.





The Golden Rule of Capital Accumulation

Golden rule savings rate

Steady-state consumption per worker:

$$c^* = f(k^*) - (n + \delta)k^*$$

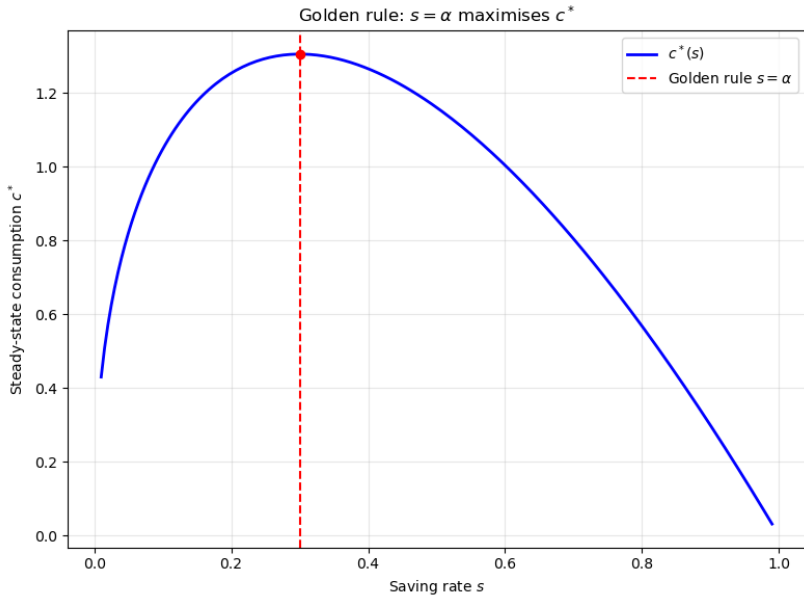
The **golden rule** k_{GR}^* maximizes c^* :

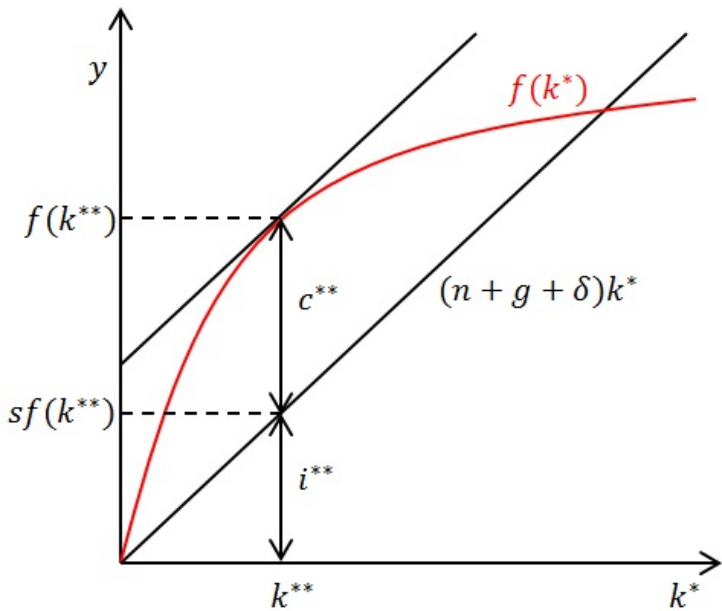
$$\frac{dc^*}{dk^*} = f'(k_{GR}^*) - (n + \delta) = 0$$

Golden rule condition

$$f'(k_{GR}^*) = n + \delta$$

For Cobb–Douglas: $\alpha k_{GR}^{*\alpha-1} = n + \delta$, so $k_{GR}^* = \left(\frac{\alpha}{n+\delta}\right)^{\frac{1}{1-\alpha}}$, which implies $s_{GR} = \alpha$.





Dynamic efficiency

- If $k^* < k_{GR}^*$ (equivalently $s < \alpha$): the economy is **dynamically efficient**.
 - Raising s increases long-run c^* but requires sacrificing current consumption.
 - There is a trade-off.
- If $k^* > k_{GR}^*$ (equivalently $s > \alpha$): the economy **over-accumulates** capital.
 - Reducing s raises consumption in *every* period—a Pareto improvement.
 - This is called **dynamic inefficiency**.

Most empirical evidence suggests real economies are dynamically efficient ($s < \alpha$).

The Discrete-Time Formulation

Capital dynamics in discrete time

Aggregate law of motion:

$$K_{t+1} = (1 - \delta)K_t + sY_t$$

Population: $L_{t+1} = (1 + n)L_t$.

Divide by L_{t+1} :

$$k_{t+1} = \frac{(1 - \delta)K_t + sF(K_t, L_t)}{(1 + n)L_t}$$

Using CRS, $F(K_t, L_t) = L_t f(k_t)$:

Discrete-time Solow equation

$$k_{t+1} = \frac{(1 - \delta)k_t + sf(k_t)}{1 + n}$$

Steady state in discrete time

At steady state $k_{t+1} = k_t = k^*$:

$$k^*(1 + n) = (1 - \delta)k^* + sf(k^*)$$

$$\Rightarrow sf(k^*) = (n + \delta)k^*$$

The same condition as in continuous time. For Cobb–Douglas:

$$k^* = \left(\frac{s}{n + \delta} \right)^{\frac{1}{1-\alpha}}, \quad y^* = \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

Numerical example

Set $(s, \alpha, \delta, n) = (0.36, 0.36, 0.04, 0.01)$.

$$k^* = \left(\frac{0.36}{0.05} \right)^{\frac{1}{0.64}} = 7.2^{1.5625} \approx 21.86$$

Start from $k_0 = 0.5$. Use $k_{t+1} = \frac{(1-\delta)k_t + sk_t^\alpha}{1+n}$ iteratively.

Adding Technological Progress

Without technological progress:

- The economy converges to a steady state with **zero growth** in per capita output.
- This contradicts the data: per capita GDP grows persistently (Kaldor SF 1).

We need to introduce **technological progress** to generate sustained per capita growth.

Re-introduce technology as **labor-augmenting** (Harrod-neutral):

$$Y = F(K, AL) = K^\alpha (AL)^{1-\alpha}$$

where A is the level of technology growing at exogenous rate $g > 0$:

$$\frac{\dot{A}}{A} = g \quad \Rightarrow \quad A(t) = A_0 e^{gt}$$

This is equivalent to $Y = zK^\alpha L^{1-\alpha}$ with $z = A^{1-\alpha}$.

TFP growth is **exogenous**: the model does not explain where technology comes from.

Effective labor and new intensive variables

In the balanced growth path (BGP), all per capita variables grow at rate g . We need variables that are **constant** in BGP.

Define per **effective worker** variables:

$$\tilde{y} \equiv \frac{Y}{AL} = \frac{y}{A}, \quad \tilde{k} \equiv \frac{K}{AL} = \frac{k}{A}$$

Then $\tilde{y} = \tilde{k}^\alpha$, and the fundamental equation becomes:

Solow equation with technology

$$\dot{\tilde{k}} = s\tilde{k}^\alpha - (n + g + \delta)\tilde{k}$$

Derivation of $\dot{\tilde{k}}$

Since $\tilde{k} = K/(AL)$:

$$\begin{aligned}\frac{\dot{\tilde{k}}}{\tilde{k}} &= \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{L}}{L} = \frac{sY - \delta K}{K} - g - n \\ &= s\frac{Y}{K} - \delta - g - n = s\frac{\tilde{y}}{\tilde{k}} - (n + g + \delta)\end{aligned}$$

Therefore:

$$\dot{\tilde{k}} = s\tilde{y} - (n + g + \delta)\tilde{k} = sf(\tilde{k}) - (n + g + \delta)\tilde{k}$$

Same structure as before, with n replaced by $n + g$.

Steady state with technology

Set $\dot{\tilde{k}} = 0$:

$$s\tilde{k}^{*\alpha} = (n + g + \delta)\tilde{k}^*$$

$$\tilde{k}^* = \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}}, \quad \tilde{y}^* = \left(\frac{s}{n + g + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

Converting back to per worker variables:

$$k(t) = A(t) \cdot \tilde{k}^* = A_0 e^{gt} \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}}$$

$$y(t) = A(t) \cdot \tilde{y}^* = A_0 e^{gt} \left(\frac{s}{n + g + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

On the BGP, y , k , and c all grow at rate g .

Balanced growth path

On the BGP, all per capita variables grow at rate g :

$$\frac{\dot{y}}{y} = \frac{\dot{k}}{k} = \frac{\dot{c}}{c} = g$$

To verify: from $y = k^\alpha A^{1-\alpha}$, take logs and differentiate:

$$g_y = \alpha g_k + (1 - \alpha)g_A$$

On the BGP, $g_k = g_y$ (since K/Y is constant by Kaldor SF 4), so:

$$g_y = \alpha g_y + (1 - \alpha)g \quad \Rightarrow \quad g_y = g = g_A$$

All aggregate variables (Y , K) grow at rate $n + g$.

The Solow model matches Kaldor's facts

On the balanced growth path:

- ① Output per worker $y = Y/L$ grows at constant rate g . ✓ (SF 1)
- ② Capital per worker $k = K/L$ grows at constant rate g . ✓ (SF 2)
- ③ Interest rate $r = f'(k) = \alpha \tilde{k}^{*\alpha-1}$ is constant. ✓ (SF 3)
- ④ Capital-output ratio $K/Y = \tilde{k}^*/\tilde{y}^*$ is constant. ✓ (SF 4)
- ⑤ Factor shares: capital share $= \alpha$, labor share $= 1 - \alpha$ (always, with Cobb–Douglas). ✓ (SF 5)

The Solow model with technological progress fits the growth facts well.
But: the **source of growth**, technological progress, is left unexplained.

Golden rule with technology

Steady-state consumption per effective worker:

$$\tilde{c}^* = f(\tilde{k}^*) - (n + g + \delta)\tilde{k}^*$$

Golden rule:

$$f'(\tilde{k}_{GR}^*) = n + g + \delta$$

For Cobb–Douglas: $s_{GR} = \alpha$ (same as before).

- If $s < \alpha$: raising s increases long-run consumption per effective worker.
- If $s > \alpha$: over-accumulation—reducing s is a Pareto improvement.

The Convergence Hypothesis

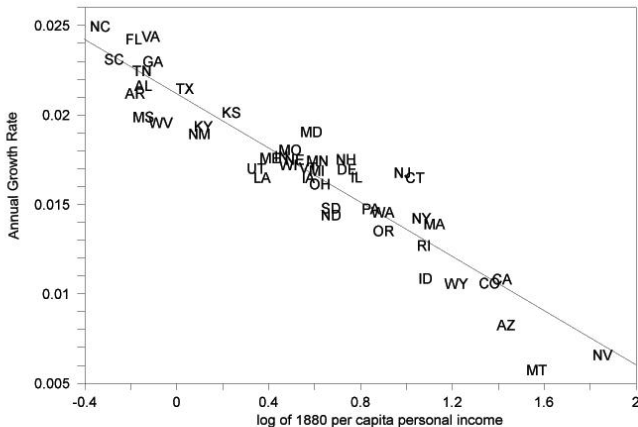
Prediction: if countries share the same s, n, δ, α, g , then they converge to the same steady state. Countries with lower initial k grow faster.

Absolute convergence: poor countries should grow faster than rich countries, unconditionally.

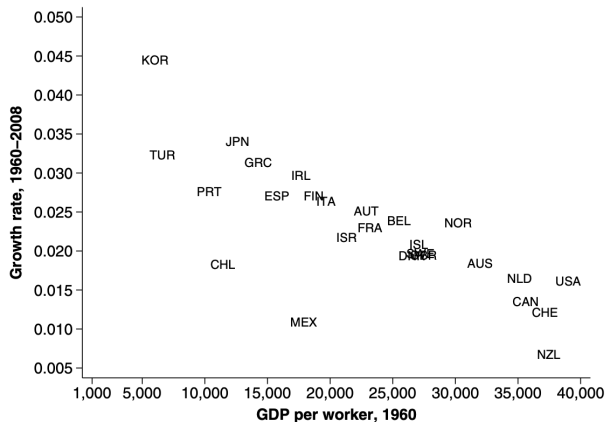
- Supported among **industrialized countries** (OECD): similar structural parameters.
- **Not** supported in the full cross-section of countries.

Convergence among US states

Figure 4: Convergence of Personal Income across U.S. States.
1880 Personal Income and Income Growth from 1880 to 1990

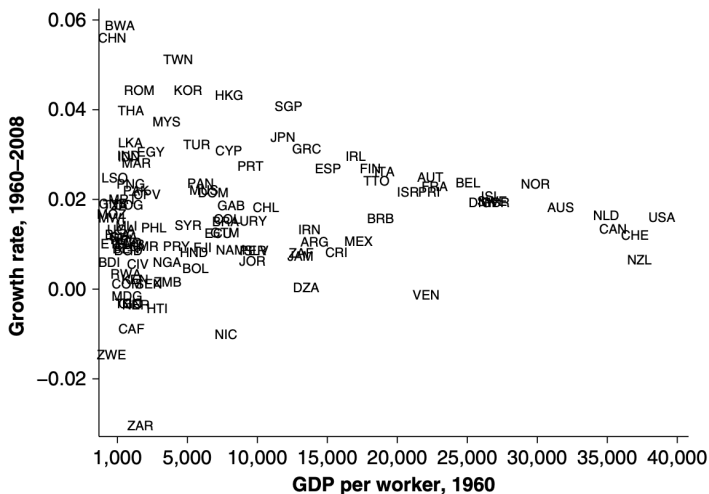


Convergence among OECD countries



SOURCE: Penn World Tables Mark 7.0 and Summers and Heston (1991)

No convergence in the full sample



SOURCE: Penn World Tables Mark 7.0 and Summers and Heston (1991)

Conditional convergence

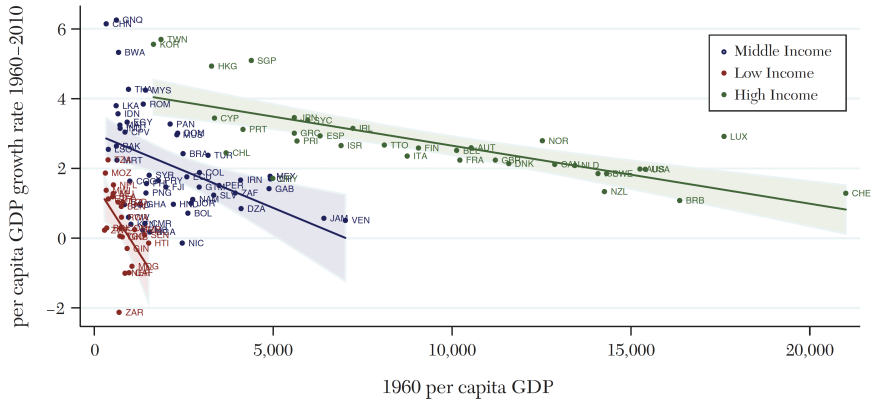
Countries differ in s , n , δ , α , so they have **different steady states**.

Conditional convergence: countries converge to *their own* steady state. Countries farther below *their own* BGP grow faster.

- Controlling for determinants of steady state (investment rate, population growth, education), there is a robust negative partial correlation between initial income and growth.
- This is the empirically relevant notion of convergence.

Capital should flow from rich (high K , low MPK) to poor (low K , high MPK) countries, speeding up this process.

(by income group)



Speed of convergence

Linearize the Solow equation around $\tilde{k}^*$. Let $\hat{k} \equiv \ln \tilde{k} - \ln \tilde{k}^*$:

$$\dot{\hat{k}} \approx -\lambda \hat{k}, \quad \lambda = (1 - \alpha)(n + g + \delta)$$

where λ is the **speed of convergence**. The half-life is $t_{1/2} = \ln 2 / \lambda$.

Calibration: $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$:

$$\lambda = 0.67 \times 0.08 = 0.053 \quad \Rightarrow \quad t_{1/2} \approx 13 \text{ years}$$

Empirical estimates: $\lambda \approx 0.02$, $t_{1/2} \approx 35$ years (slower than the basic model predicts). This is one motivation for augmented Solow models (e.g., Mankiw–Romer–Weil 1992 with human capital, where effective α is higher).

Solow Model with Human Capital

(Mankiw–Romer–Weil, 1992)

Motivation: the convergence speed puzzle

The basic Solow model with $\alpha = 1/3$ predicts $\lambda \approx 0.05$ (half-life ≈ 13 years).

Empirical evidence: $\lambda \approx 0.02$ (half-life ≈ 35 years). The model predicts **too-fast convergence**.

Diagnosis: $\lambda = (1 - \alpha)(n + g + \delta)$. To slow convergence, we need a higher effective capital share α .

Solution (Mankiw, Romer, and Weil 1992, *QJE*): introduce **human capital** as a second accumulable factor. This raises the effective capital share and slows convergence.

Production with human capital

Production:

$$Y(t) = K(t)^{\alpha_K} H(t)^{\alpha_H} [A(t)L(t)]^{1-\alpha_K-\alpha_H}$$

with $0 < \alpha_K + \alpha_H < 1$. Here $H(t)$ is aggregate human capital.
In per effective worker terms ($k = K/(AL)$, $h = H/(AL)$):

$$y(t) = k(t)^{\alpha_K} h(t)^{\alpha_H}$$

Laws of motion: households save fractions s_K and s_H of income for physical and human capital respectively:

$$\dot{k}(t) = s_K y(t) - (\delta_K + n + g) k(t)$$

$$\dot{h}(t) = s_H y(t) - (\delta_H + n + g) h(t)$$

We now have a system of **two** differential equations.

Steady state

At steady state $\dot{k} = 0$ and $\dot{h} = 0$:

$$s_K(k^*)^{\alpha_K}(h^*)^{\alpha_H} = (\delta_K + n + g)k^*$$

$$s_H(k^*)^{\alpha_K}(h^*)^{\alpha_H} = (\delta_H + n + g)h^*$$

Solving (assuming $\delta_K = \delta_H = \delta$ for simplicity):

$$k^* = \left[\left(\frac{s_K}{\delta + n + g} \right)^{1-\alpha_H} \left(\frac{s_H}{\delta + n + g} \right)^{\alpha_H} \right]^{\frac{1}{1-\alpha_K-\alpha_H}}$$

$$h^* = \left[\left(\frac{s_K}{\delta + n + g} \right)^{\alpha_K} \left(\frac{s_H}{\delta + n + g} \right)^{1-\alpha_K} \right]^{\frac{1}{1-\alpha_K-\alpha_H}}$$

Implications for convergence

The effective capital share is now $\alpha_K + \alpha_H$, not just α_K .

Speed of convergence becomes:

$$\lambda = (1 - \alpha_K - \alpha_H)(n + g + \delta)$$

With $\alpha_K = 1/3$ and $\alpha_H = 1/3$: $\alpha_K + \alpha_H = 2/3$, so

$$\lambda = (1/3)(0.08) \approx 0.027 \quad \Rightarrow \quad t_{1/2} \approx 26 \text{ years}$$

Much closer to the empirical estimate of $\lambda \approx 0.02$.

The augmented model also does better at explaining cross-country income differences: steady-state income depends on s_K , s_H , and n , giving richer variation.

Phase diagram with human capital

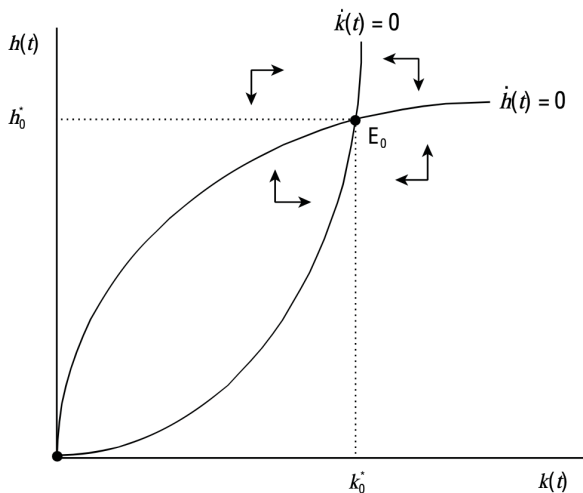


Figure 13.6: Augmented Solow-Swan model

Growth Accounting

The Solow residual

From $Y = A \cdot K^\alpha L^{1-\alpha}$, take logs and differentiate:

$$\frac{\dot{Y}}{Y} = \frac{\dot{A}}{A} + \alpha \frac{\dot{K}}{K} + (1 - \alpha) \frac{\dot{L}}{L}$$

Or in growth rates:

$$g_Y = g_A + \alpha g_K + (1 - \alpha) g_L$$

The **Solow residual** is the part of output growth not explained by factor accumulation:

$$g_A = g_Y - \alpha g_K - (1 - \alpha) g_L$$

Also called **Total Factor Productivity (TFP)** growth. In practice, g_A accounts for a large fraction of growth—this is the part that the Solow model leaves unexplained.

Growth accounting: interpretation

- Inputs K and L are directly measurable.
- Factor shares α and $1 - \alpha$ can be estimated from national accounts.
- TFP growth g_A is a **residual**—a “measure of our ignorance.”

Empirical finding: TFP growth typically accounts for 50–70% of per capita output growth in developed countries. This highlights the central role of technology, which the Solow model takes as exogenous.

TFP (labor-augmenting, US = 1)

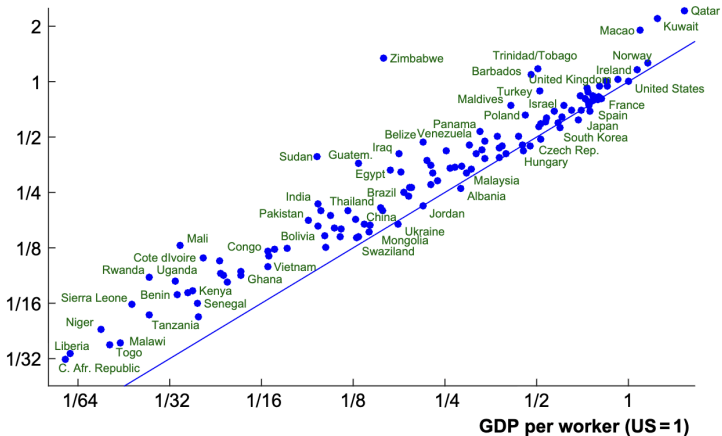


Fig. 29 Total factor productivity, 2010. Source: *Computed using the Penn World Tables 8.0 assuming a common value of $\alpha = 1/3$.*

Evaluation and What Comes Next

Strengths of the Solow model

- Simple and transparent framework for understanding growth.
- Clear predictions about the role of s , n , g , δ in long-run income levels.
- Transition dynamics and conditional convergence.
- Matches Kaldor's stylized facts on the BGP.
- Foundation for growth accounting.

Limitations of the Solow model

- Savings rate s is **exogenous**: no optimizing behavior by households.
 - Cannot analyze welfare or optimal policy.
 - Cannot study how s responds to changes in the environment.
- Technology growth g is **exogenous**: the engine of long-run growth is a black box.
- Limited role for policy: only level effects, not growth effects.
- Predicts too-fast convergence (without human capital augmentation).

Bridge to Ramsey–Cass–Koopmans

- The Solow model pins long-run levels and transition speed, but not optimizing behavior.
- Next step: **endogenize** the consumption-saving decision.
- The Ramsey–Cass–Koopmans model replaces the exogenous s with an Euler equation derived from household optimization (as in Lecture #1).
- Core law of motion remains $\dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n + g + \delta)\tilde{k}$, but now $\tilde{c}$ is chosen optimally.

Exercise: comparative statics in the Solow model

Consider a Solow economy with $f(k) = k^\alpha$, $\alpha = 1/3$, $s = 0.2$, $n = 0.02$, $\delta = 0.05$, $g = 0.02$.

- 1 Compute the steady-state values $\tilde{k}^*$, $\tilde{y}^*$, and $\tilde{c}^*$.
- 2 Compute the golden rule savings rate s_{GR} and the golden rule capital $\tilde{k}_{GR}^*$.
- 3 Suppose s increases permanently to $s' = 0.3$. Find the new steady state. Describe the transition path of $\tilde{k}$, $\tilde{y}$, and $\tilde{c}$ (qualitatively).
- 4 Compute the speed of convergence λ and the half-life $t_{1/2}$.
- 5 Is this economy dynamically efficient? Explain.

► Solution

Exercise: growth accounting

Suppose in a country over a 10-year period:

- Output grew at 5% per year
- Capital stock grew at 6% per year
- Labor force grew at 2% per year
- Capital share $\alpha = 0.35$

- 1 Compute TFP growth g_A .
- 2 What fraction of output growth is explained by TFP?
- 3 If we interpret A as labor-augmenting technology ($Y = K^\alpha (AL)^{1-\alpha}$), what is the growth rate of A ?

APPENDIX

Solution: comparative statics exercise

1. $\tilde{k}^* = \left(\frac{0.2}{0.09}\right)^{3/2} = 2.222^{1.5} \approx 3.31$, $\tilde{y}^* = 3.31^{1/3} \approx 1.49$,
 $\tilde{c}^* = 0.8 \times 1.49 \approx 1.19$.
2. $s_{GR} = \alpha = 1/3 \approx 0.333$. $\tilde{k}_{GR}^* = \left(\frac{1/3}{0.09}\right)^{3/2} = 3.704^{1.5} \approx 7.13$.
3. New: $\tilde{k}^{*'} = \left(\frac{0.3}{0.09}\right)^{3/2} = 3.333^{1.5} \approx 6.09$. During transition, $\tilde{k}$ rises gradually from 3.31 to 6.09; $\tilde{y}$ rises; $\tilde{c}$ initially *drops* (higher s , lower $1 - s$) then rises as $\tilde{y}$ increases.
4. $\lambda = (1 - 1/3)(0.09) = 0.06$. $t_{1/2} = \ln 2 / 0.06 \approx 11.6$ years.
5. $s = 0.2 < \alpha = 1/3 = s_{GR}$, so the economy is dynamically efficient (below the golden rule).

Further Readings

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